System Design using Verilog

GMR Institute of Technology

Contents of the lecture

(1) “if….else” statement

(2) Write behaviour model of 4 to 1 multiplexer using “if….else” statement

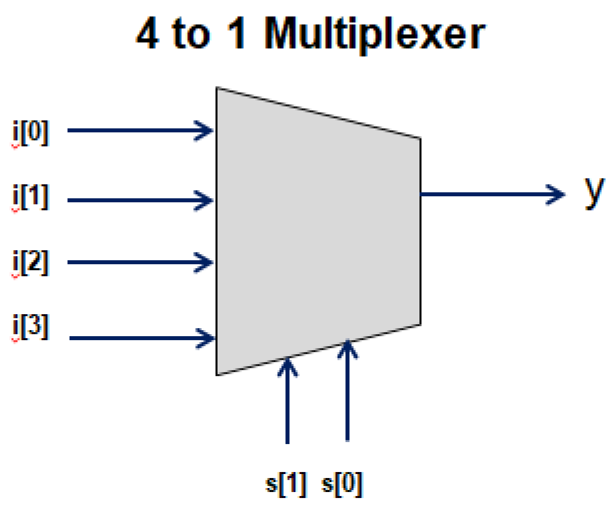
(3) Test bench of 4 to 1 multiplexer

(4) Software Demonstration

GMR Institute of Technology

Example:
Write behaviour model of 4 to 1 multiplexer

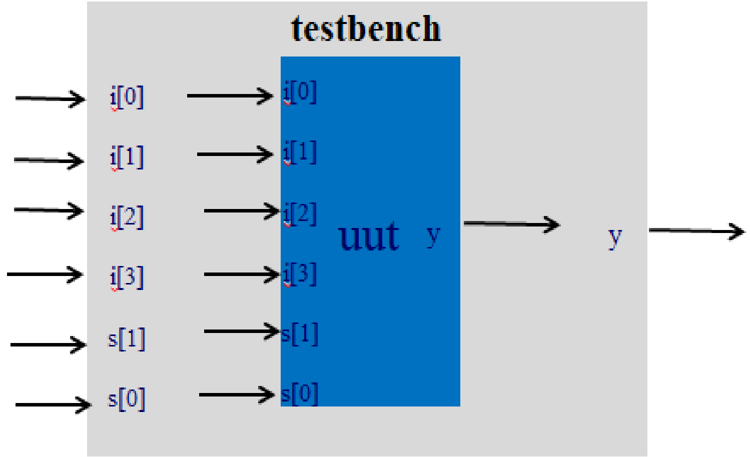
```
module mux4to1(i,s,y);
input [0:3]i;
input [1:0]s;
output y;
reg y;
always @(i,s)
if (s==2'b00)
y=i[0];
else if (s==2'b01)
y=i[1];
else if (s==2'b10)
y=i[2];
else if (s==2'b11)
y=i[3];
endmodule
```



S[1]	s[0]	y
0	0	i[0]
0	1	i[1]
1	0	i[2]
1	1	i[3]

Test Bench of 4 to 1 Mux

```
module mux4to1 (i,s,y);
input [0:3]i;
input [1:0]s;
output y;
reg y;
always @(i,s)
if (s==2'b00)
y=i[0];
else if (s==2'b01)
y=i[1];
else if (s==2'b10)
y=i[2];
else if (s==2'b11)
y=i[3];
endmodule
```




```
module testbench;
reg [0:3]i;reg [1:0]s; wire y;
mux4to1 uut (.i(i),.s(s),.y(y));
initial
begin
$monitor($time,"i[0]=%b,i[1]=%b, i[2]=%b,i[3]=%b,
s[1]=%b,s[0]=%b,y=%b",i[0],i[1],i[2],i[3],s[1],s[0],y);
#5 i=4'b0101;s=2'b00;
#5 i=4'b0101;s=2'b01;
#5 i=4'b0101;s=2'b10;
#5 i=4'b0101;s=2'b11;
#5 i=4'b1100;s=2'b00;
#5 i=4'b1100;s=2'b01;
#5 i=4'b1100;s=2'b10;
#5 i=4'b1100;s=2'b11;
#5 $finish;
end
endmodule
```

Lecture 25
Structure Modeling (2 to 1 Multiplexer)

Contents of the lecture

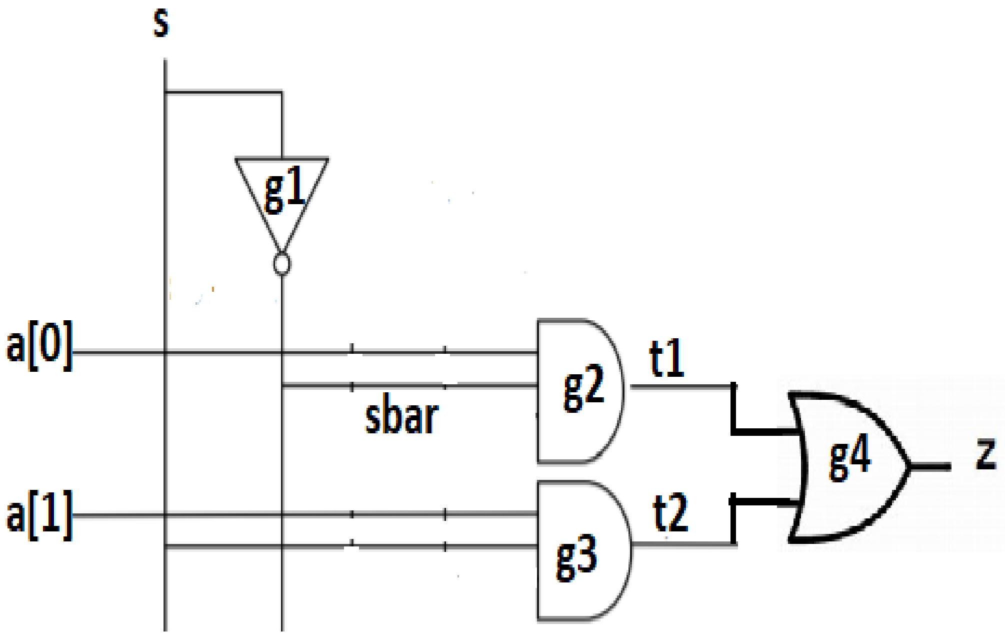
(1) Structure model of 2-to-1 Multiplexer

(2) Test bench of 2-to-1 Multiplexer

(3) Software Demonstration

Structure model of 2-to-1 Multiplexer

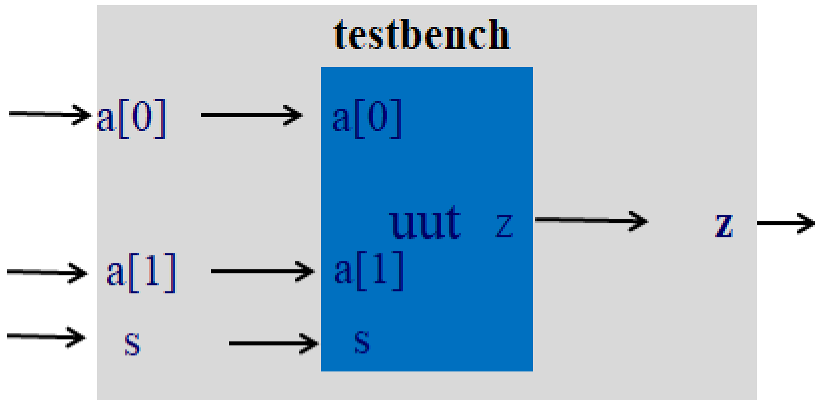
```
module mux_2_1 (a,s,z);
input [0:1]a;
input s;
output z;
wire sbar, t1, t2;
not g1 (sbar, s);
and g2 (t1, a[0], sbar);
and g3 (t2, a[1], s);
or g4 (z, t1, t2);
endmodule
```

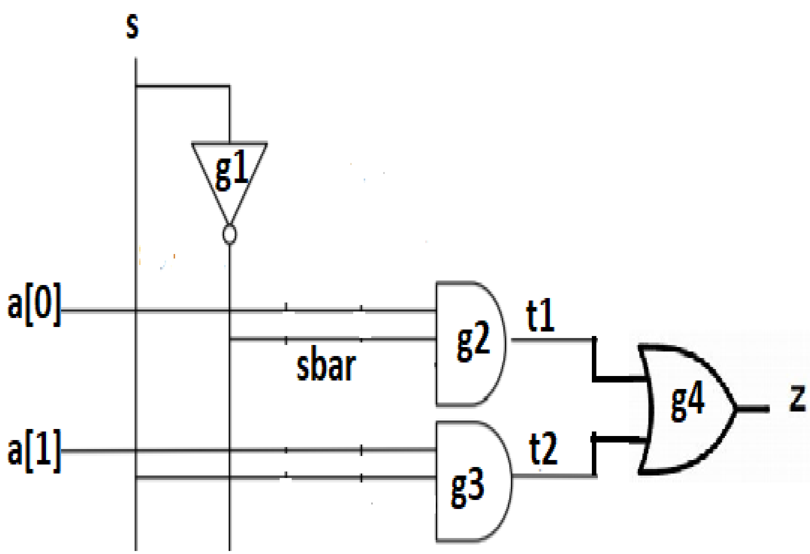


Testbench of 2-to-1 Multiplexer

```
module mux_2_1 (a,s,z);
input [0:1]a;
input s;
output z;
wire sbar, t1, t2;
not g1 (sbar, s);
and g2 (t1, a[0], sbar);
and g3 (t2, a[1], s);
or g4 (z, t1, t2);
endmodule
```

```
module testbench;
reg [0:1]a;reg s; wire z;
mux_2_1 uut (.a(a),.s(s),.z(z));
initial
begin
$monitor($time,"a[0]=%b,a[1]=%b,s=%b,z=%b",a[0],a[1]
,s,z);
#5 a=2'b01;s=0;
#5 a=2'b01;s=1;
#5 a=2'b10;s=0;
#5 a=2'b10;s=1;
#5 $finish;
end
endmodule
```





Software Demonstration

Contents of the lecture

(1) Write behaviour model of 2 to 4 Decoder using "if….else" statement

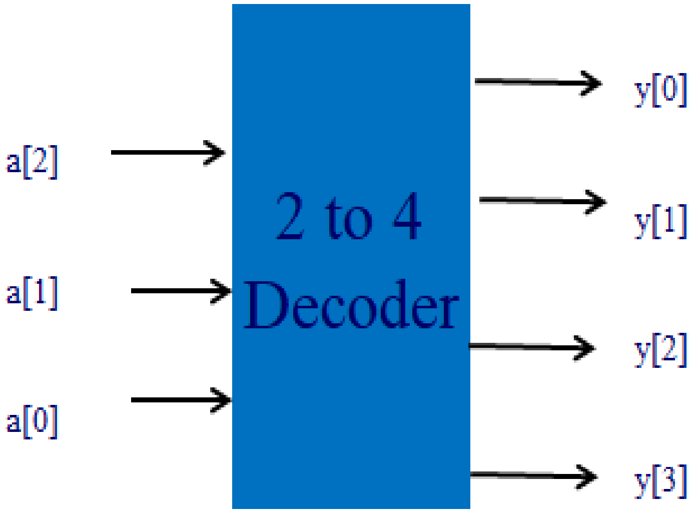
(2) Test bench of 2 to 4 Decoder

(3) Software Demonstration

Example:

Write behavior model of 2 to 4 Decoder

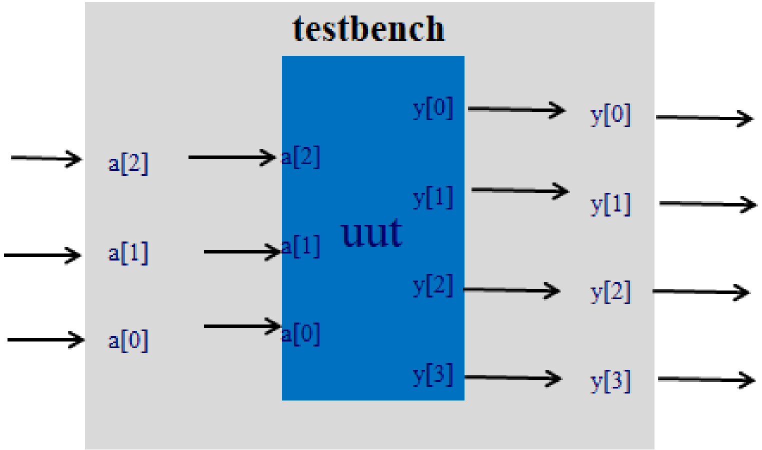
```
module decoder_2_to_4 (a,y);
input [2:0]a;
output [0:3]y;
reg [0:3]y;
always @(a)
if (a==3'b100)
y= 4'b1000;
else if (a==3'b101)
y= 4'b0100;
else if (a==3'b110)
y= 4'b0010;
else if (a==3'b111)
y= 4'b0001;
else
y= 4'b0000;
endmodule
```



a[2]	a[1]	a[0]	y[0]	y[1]	y[2]	y[3]
0	x	x	0	0	0	0
1	0	0	1	0	0	0
1	0	1	0	1	0	0
1	1	0	0	0	1	0
1	1	1	0	0	0	1

Test Bench of 2 to 4 Decoder

```
module  decoder_2_to_4 (a,y);
input  [2:0]a;
output  [0:3]y;
reg [0:3]y;
always  @(a)
if (a==3'b100)
y= 4'b1000;
else if (a==3'b101)
y= 4'b0100;
else if (a==3'b110)
y= 4'b0010;
else if (a==3'b111)
y= 4'b0001;
else
y= 4'b0000;
endmodule
```




```
module testbench;
reg [2:0]a; wire [0:3]y;
decoder_2_to_4 uut (.a(a),.y(y));
initial
begin
$monitor($time,"a[2]=%b,a[1]=%b,
a[0]=%b,y[0]=%b,y[1]=%b,y[2]=%b, ,y[3]=%b",
a[2],a[1], a[0],y[0],y[1],y[2],y[3]);
#5 a=3'b100;
#5 a=3'b101;
#5 a=3'b110;
#5 a=3'b111;
#5 a=3'b000;
#5 a=3'b101;
#5 $finish;
end
endmodule
```

GMR Institute of Technology

Software Demonstration